

Shared evidence • Plant-system cards

The plant-system service desk

Name _____ Date _____

Use these biological function cards. The service cases are fictional; do not damage plants.

A • Roots

Anchor the plant. Absorb water and dissolved mineral nutrients.

C • Green leaves

Use light energy, water and carbon dioxide from air to make sugars.

E • Service case

A plant's leaves are shaded by a box. Its roots still have water; the box prevents light reaching the leaves.

B • Stem

Supports leaves and flowers. Contains pathways transporting water and other substances.

D • Flowers

Take part in reproduction. After successful pollination and fertilisation, seeds can form.

F • Connection case

Water has entered the roots. It still needs a pathway through the stem to reach the leaves.

W1. Choose two parts and explain how their jobs connect.

W2. Which input is missing in E? Why will extra minerals not replace it?

Route 1 • Match useful jobs

Word bank: roots / stem / leaves / flowers.

S1. Which part absorbs water from soil?

S2. Which part connects roots to leaves and transports water?

S3. Which green parts make sugars using light?

S4. Which parts take part in making seeds?

S5. Mineral nutrients are...

- A. Needed substances taken in with water.
- B. Ready-made sugar meals eaten from soil.

Route 2 • Write the service guide

Use a part name, a function and a reason.

M1. Explain how roots and stem help water reach a leaf.

M2. Explain why E needs light as well as water.

M3. Give two jobs of the stem.

M4. Correct “flowers only make the plant look colourful”.

Route 3 • Review a proposed repair

A pupil says: “The shaded plant has water, so giving it twice as many minerals will replace the missing light.”

T1. Explain the mistake using card C.

T2. Propose a repair that addresses the missing input.

T3. Explain why roots alone cannot account for water reaching leaves.

T4. Must every flower become a seed-bearing fruit? Use a careful sentence.

Exit and reflection

Point, draw, write or tell an adult.

E1. Which part absorbs water from soil?

E2. Where do green plants make sugars using light?

E3. What do flowers help the plant do?

R1. Which idea did you improve or defend? What evidence helped?

R2. What useful question would you investigate next?
